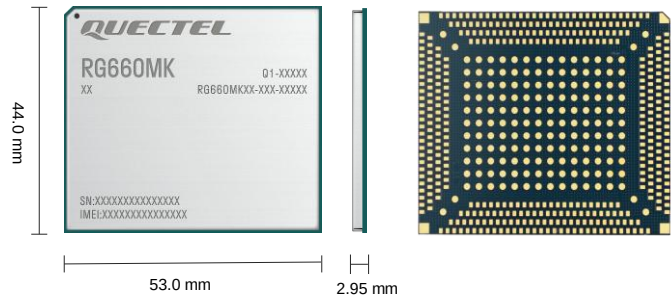




Quectel RG660MK-EU

IoT/ eMBB/ uRLLC-Optimized 5G Sub-6 GHz LGA Module



Quectel RG660MK-EU is a 5G Sub-6 GHz LGA module optimized specially for IoT/ eMBB/ uRLLC applications. Adopting the 3GPP Rel-18 technology, the module supports 3TX/ 8RX, L4S and Wi-Fi 8. The module supports the 5G modes of both NSA and SA and the deployment of Option 3x, 3a, 3, and 2, and is backward compatible with 4G.

RG660MK-EU is an industrial-grade module for industrial and commercial applications only.

The module integrates a multi-constellation GNSS receiver that supports GPS, GLONASS, BDS, Galileo and QZSS. It helps to simplify the product design and can achieve high-precision, fast, and dependable positioning.

RG660MK-EU has a quad-core 2.34 GHz CPU and integrates network and VPN accelerators, which extends the applicability of the module to a wide range of eMBB, IoT and uRLLC applications such as 5G router, CPE, MHS, industrial router, and home gateway.



Key Features

- ✓ Sub-6 GHz 5G module with LGA form factor, optimized for IoT/ eMBB/ uRLLC applications
- ✓ Both NSA and SA modes supported
- ✓ 5G 6CA supported
- ✓ 3TX/ 8RX, L4S, Wi-Fi 8 supported
- ✓ Multi-mode network coverage, compatible with 4G network (Cat 20)
- ✓ Transmission Mode 9 (TM9) supported, increased spectral efficiency and improved performance
- ✓ Multi-constellation GNSS receiver available for applications requiring fast and accurate fixes in any environment

 5G Sub-6 GHz Bands Supported	 Max. 2.0 Gbps (DL) Max. 211 Mbps (UL)	 Multi-Constellation GNSS
 Embedded Abundant Protocols	 LGA Form Factor	 USB 3.2/ PCIe 4.0 Super Speed Interface
 VoLTE/ VoNR (Optional)	 Quectel Enhanced API	 Wi-Fi 8
 3TX/ 8RX	 L4S	

Version: 1.0.1 | Status: Preliminary

Quectel RG660MK-EU*

5G Sub-6		RG660MK-EU
Region/ Operator		EMEA/ APAC ^① / Brazil
Dimensions (mm)		53.0 × 44.0 × 2.95
Temperature Range		
Operation Temperature		-30 °C to +70 °C
Extended Temperature		-40 °C to +85 °C
Frequency Band		
5G	NR NSA	n1/ 3/ 5 / 7/ 8/ 20/ 28/ 38/ 40/ 41/ 75/ 76/ 77/ 78/ 79
	NR SA	n1/ 3/ 5/ 7/ 8/ 20/ 28/ 38/ 40/ 41/ 75/ 76/ 77/ 78/ 79
	DL CA	DL 6CA/ UL 2CA
	8RX	n41/ 77/ 78/ 79
	3TX ^②	FDD: n1/ 3/ 7/ 8/ 28
		TDD: n77/ 78/ 79
	DL 4 × 4 MIMO	n1/ 3/ 7/ 8/ 20/ 28/ 38/ 40/ 41/ 75/ 76/ 77/ 78/ 79
	UL 2 × 2 MIMO	n1/ 3/ 7/20/ 28/ 38/ 40/ 41/ 77/ 78/ 79
LTE	LTE-FDD	B1/ 3/ 5/ 7/ 8/ 18/ 19/ 20/ 26/ 28/ 32
	LTE-TDD	B38/ 40/ 41/ 42/ 43
	LAA	-
	DL 4 × 4 MIMO	B1/ 3/ 7/ 8/ 20/ 28/ 32/ 38/ 40/ 41/ 42/ 43
UMTS	WCDMA	-
GNSS		GPS/ BDS/ GLONASS/ Galileo/ QZSS, (L1 + L5) ^③
Certification		
Regulatory/ Conformance*		Global: GCF Europe: CE Australia/ New Zealand: RCM
Carrier* (Based on Carrier’s RFP)		TBD
Others*		RoHS
Data Rate (Max.) ^④		
5G SA Sub-6 GHz		9.35 Gbps (DL)/ 2.81 Gbps (UL)
5G NSA Sub-6 GHz		5.87 Gbps (DL)/ 1.46 Gbps (UL)
LTE		2.0 Gbps (DL)/ 211 Mbps (UL)
DC-HSPA+		-
Interface		
Antenna		Cellular: × 8, GNSS: × 1
(U)SIM		× 2 (DSDA*)
UART		× 3
USB 3.2/ 2.0		USB 3.2 Gen2 × 1, USB 2.0 × 1
USXGMII		× 2
PCIe		PCIe 4.0 × 1, PCIe 3.0 × 2
PCM		× 2
SPI		× 3
I2C		× 4
ADC		× 4
SDIO		× 1

Note:
1. ①: Exclude China/ Japan.
2. ②: Supports any possible combination of FDD + TDD ULMIMO bands.
3. ③: Optional.

4.④: Theoretical only. The actual values depend on network conditions.
5. *: Under development/ planning/ in progress.
6. TBD: To Be Determined.



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5G Sub-6	RG660MK-EU
Audio	
Voice (Optional)	Digital Audio and VoLTE/ VoNR
Enhanced Feature	
DTMF	●
FOTA	●
(U)SIM Card Detection	●
3TX	●
8RX	●
L4S	●
Wi-Fi 8*	●
Electrical/ RF Feature	
Power Supply	3.3–4.4 V, typical 3.8 V
Power Consumption	TBD @ Power down TBD @ Sleep TBD @ Idle (USB connected)
Transmit Power	Class 3 (23 dBm ±2 dB) for LTE-FDD bands Class 3 (23 dBm ±2 dB) for LTE-TDD bands Class 3 (23 dBm ±2 dB) for 5G NR bands Class 2 (26 dBm +2/ -3 dB) for 5G NR HPUE: n41/ 77/ 78/ 79 Class 1.5 (29 dBm +2/ -3 dB) for 5G NR HPUE: n77/ 78/ 79

Note:
1. ●: Supported.
2. TBD: To Be Determined.
3. *: Under development/ planning/ in progress.

